

Routing, Cache, Cost, and Streaming Performance A/B Study

Model: `minimax/minimax-m3`

This report compares Infron and OpenRouter using the standard benchmark matrix: 4 experiment groups, 50 rounds per group, streaming enabled, Chat Completions protocol, default reasoning behavior, and short/medium/long prompt length tiers.

Data Quality

- Completed request-level records: 3196
- Comparable A/B pairs: 799
- Excluded records: 2 total; 1 incomplete, 0 anomalous usage, 1 unequal input-token pairs.
- Request JSONL SHA256: `333491a80a3af487f260ea9b9f6b7cd3f863e3b150f025ea62796a345bf8caa0`
- Pair CSV SHA256: `04e9799be91b7b05670e88c20259466bba36f0132affa8f618fe73864239406d`

Aggregate Results

Routing objective	Provider	Effective rounds	Cache hit rate	Total cost (USD)	Avg latency (ms)	P95 latency (ms)	Avg TTFT (ms)	P95 TTFT (ms)	Output TPS
throughput	infron	199	99.75%	\$0.44453200	4,187.992	6,968.981	3,906.654	6,495.255	3.820
throughput	openrouter	199	97.78%	\$0.42616452	3,520.482	5,479.748	3,320.929	5,200.601	4.480
price	infron	200	87.07%	\$0.86923800	4,215.047	7,121.548	3,970.207	6,674.128	5.706
price	openrouter	200	99.71%	\$0.38910666	3,183.654	4,902.943	2,988.125	4,725.499	5.026
latency	infron	200	99.56%	\$0.38928600	3,610.426	5,793.898	3,345.935	5,363.684	4.432
latency	openrouter	200	97.80%	\$0.50588802	3,322.030	5,299.352	3,069.540	4,975.527	4.816
ttft	infron	200	99.56%	\$0.38928600	3,405.387	5,775.962	3,123.363	5,550.827	4.698
ttft	openrouter	200	99.71%	\$0.49761108	3,616.816	5,975.508	3,377.282	5,208.558	4.424

Readout

- `throughput` : cache winner Infron; cost winner OpenRouter; latency winner OpenRouter; TTFT winner OpenRouter; throughput winner OpenRouter.
- `price` : cache winner OpenRouter; cost winner OpenRouter; latency winner OpenRouter; TTFT winner OpenRouter; throughput winner Infron.

- `latency` : cache winner Infron; cost winner Infron; latency winner OpenRouter; TTFT winner OpenRouter; throughput winner OpenRouter.
- `ttft` : cache winner OpenRouter; cost winner Infron; latency winner Infron; TTFT winner Infron; throughput winner Infron.

Overall, this run shows no single platform dominating every axis. Infron is strongest on cache reuse in three of four routing objectives and has lower cost in latency/TTFT modes. OpenRouter is faster in throughput, price, and latency modes, while Infron has the lower TTFT in the explicit TTFT objective. The price objective is the largest outlier for Infron cost because cache reuse was materially lower than OpenRouter in that group.

Source Artifacts

- Run directory: `export/minimax_m3_all_experiments/
routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_2026070`
- Raw records: `export/minimax_m3_all_experiments/
routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_2026070
records.json`
- Request corpus: `export/minimax_m3_all_experiments/
routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_2026070
benchmark_requests.jsonl`
- Pair CSV: `export/minimax_m3_all_experiments/
routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_2026070
benchmark_pairs.csv`
- Summary: `export/minimax_m3_all_experiments/
routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_2026070
summary.json`